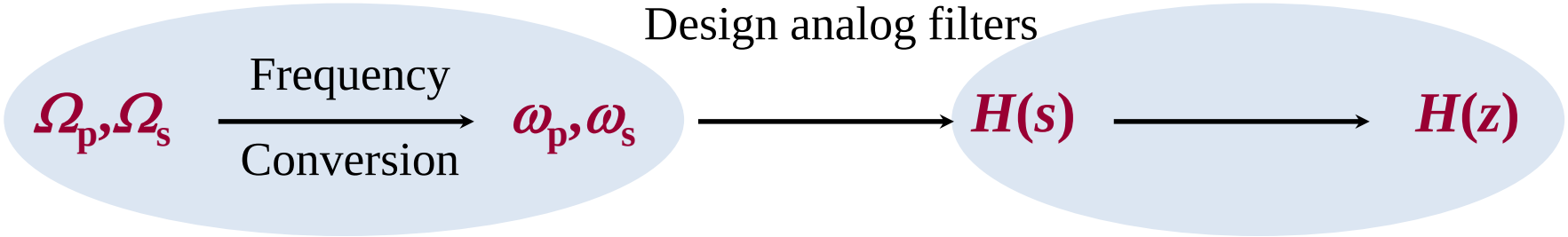
Digital Signal Processing

Zhou Hang

Impulse invariance method

- Question
- Design principle
- Design steps

Question



How to convert an analog filter into a digital filter?

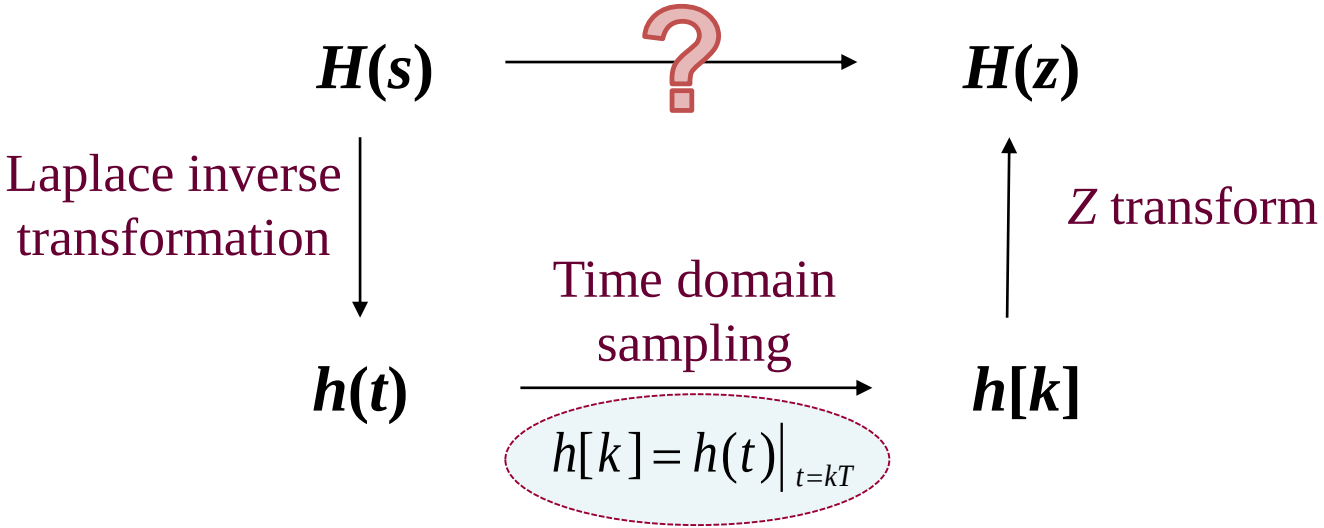
Method 1: Pulse Response Invariant Method

Method 2: Bilinear Transformation Method

Design Principle

Time domain approximation idea:

Simulate the unit impulse response $h(t)$ of AF using the unit impulse response $h[k]$ of DF.



Design Principle

$$H(s)=\sum_{i=1}^{M}\frac{A_i}{s-p_i} \quad \Longrightarrow \quad H(z)=\sum_{i=1}^{M}\frac{A_i}{1-e^{p_iT}z^{-1}}$$

AF pole $s_i = p_i$ —Mapping→ DF pole $z_i = e^{p_iT}$

The mapping relationship between AF poles and DF poles is:

$$z_i = e^{s_iT}$$

Design Principle

When transforming $H(s)$ to $H(z)$, the mapping from the analog domain to the digital domain must meet the following conditions:

(1) The frequency characteristics remain invariable, that is, the jω axis of the s-plane is mapped to the unit circle of the z-plane;

(2) The transformed filter is still stable, that is, the left half plane of s is mapped to the unit circle of the z-plane.

The mapping relationship between AF poles and DF poles is:

If $s_i = \sigma_i + j\omega_i$, then:

$$z_i = e^{s_i T}$$

$$z_i = e^{\sigma_i T} e^{j\omega_i T}$$

$$|z_i| = e^{\sigma_i T}$$

Design Principle

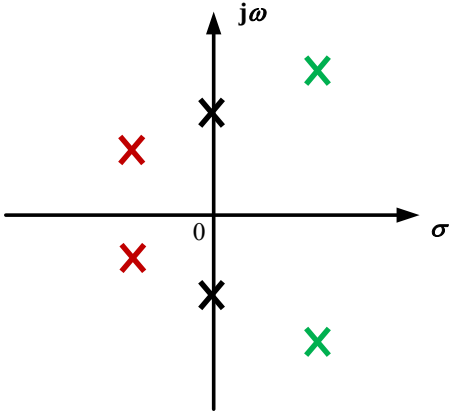
$|z_i| = e^{\sigma_i T}$

$s_i = \sigma_i + j\omega_i$

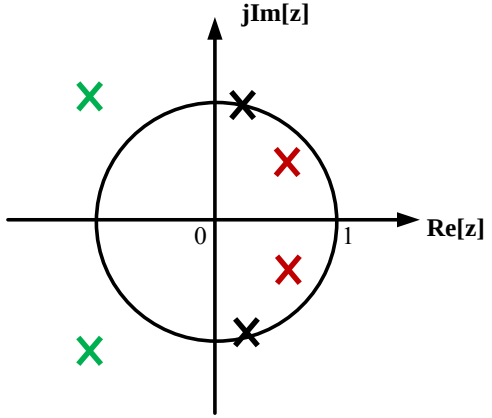
σi=0 (Located on the imaginary axis of the *s*-plane) ⟺ $|z_i|$=1 (Located on the unit circle of the *z*-plane)

σ_i<0 (Located on the left half plane of *s*) ⟺ $|z_i|$<1 (Located within the unit circle of the *z*-plane)

σ_i>0 (Located in the right half plane of *s*) ⟺ |zi|>1 (Located outside the unit circle in the *z*-plane)



Distribution of AF poles in the *s*-plane

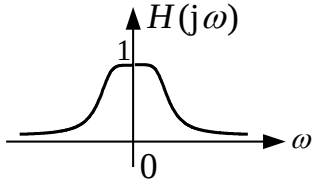


Distribution of DF poles in the *z*-plane

Design Principle

The relationship between $H(e^{j\Omega})$ and $H(j\omega)$

$$H(e^{j\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{\infty} H[j(\omega - n\omega_{sam})]$$

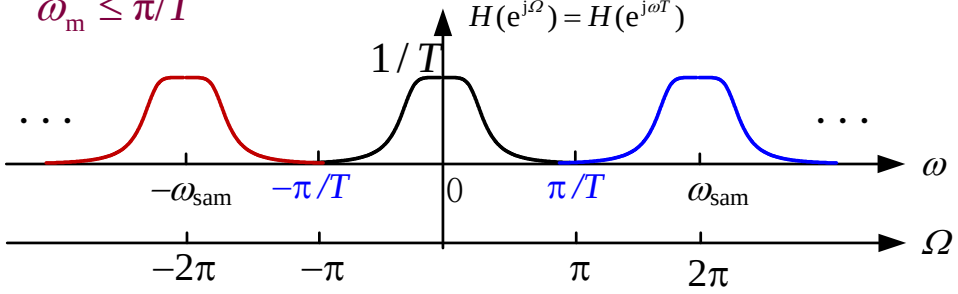


If the frequency response of AF is band limited, that is

$$H(j\omega) = 0, \quad |\omega| > \frac{\pi}{T} \qquad \omega_m \le \pi/T$$

$T \le \dfrac{\pi}{\omega_m}$, No aliasing will occur

$$H(e^{j\Omega}) = \frac{1}{T}H(j\omega), \quad |\Omega| \le \pi$$



The relationship between Ω and ω is satisfied: $\Omega = \omega T$

Design Principle

Advantages and disadvantages of pulse response invariant method

Advantages:

The frequency relationship between digital filters and analog filters is linear.

$$\Omega = \omega T$$

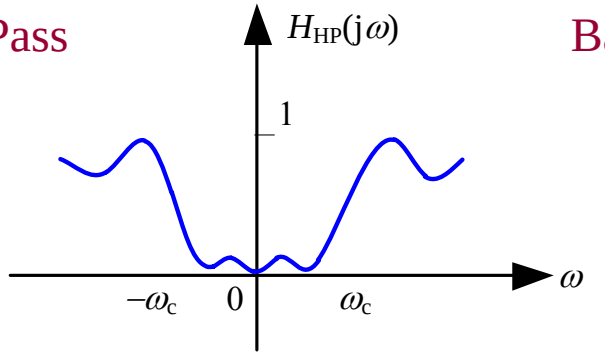
Disadvantages:

This method cannot be used to design high pass, band stop, and other filters.

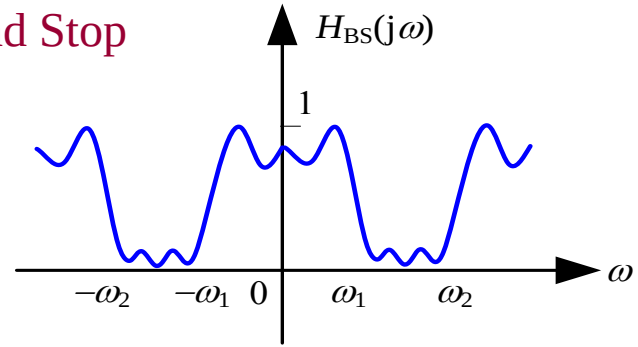


Design Principle

High Pass



Band Stop



Reason:

The high-frequency parts of high pass and band stop filters do not attenuate, resulting in severe spectral aliasing.

Design Steps

1. Convert the frequency index $\{\Omega_i\}$ of DF to the frequency index $\{\omega_i\}$ of AF

$$\omega_i = \Omega_i / T$$

2. Design the $H(s)$ of the analog filter based on its specifications.

3. Using the impulse response invariant method, convert $H(s)$ to $H(z)$.

$$\frac{1}{s - s_i} \longrightarrow \frac{1}{1 - \mathrm{e}^{s_i T} z^{-1}}$$

[E.g] **Using BW type analog low-pass filter and impulse response invariant method to design and meet the specifications Ω_p=0.2π rad, Ω_s=0.6π rad, A_p≤2dB, A_s≥15dB digital low-pass filter.**

Solution:

(1) Convert digital low-pass indicators to analog low-pass indicators

$\omega_p=\Omega_p/T=0.2\pi/T$, $\omega_s=\Omega_s/T=0.6\pi/T$, $A_p\le 2$dB, $A_s\ge 15$dB

(2) Design BW type analog low-pass filter

$$N \ge \frac{\lg\left(\frac{10^{0.1A_s}-1}{10^{0.1A_p}-1}\right)}{2\lg(\omega_s/\omega_p)} = 2 \qquad \omega_c = \frac{\omega_p}{(10^{0.1A_s}-1)^{1/(2N)}} = 0.7185/T$$

$$H_L(s) = \frac{1}{\left(\frac{s}{\omega_c}\right)^2+\sqrt{2}\frac{s}{\omega_c}+1} = \frac{0.5162}{T^2s^2+1.0161Ts+0.5162}$$

[E.g] **Using BW type analog low-pass filter and impulse response invariant method to design and meet the specifications Ω_p=0.2π rad, Ω_s=0.6π rad, A_p≤2dB, A_s≥15dB digital low-pass filter.**

Solution:

(3) Convert analog low-pass filter to digital low-pass filter

$$H_L(s)=\frac{-0.5080\,\mathrm{j}}{s-s_1}+\frac{0.5080\,\mathrm{j}}{s-s_2}$$

Poles are s_1= (–0.5080+0.5080j)/T, s_2=(–0.5080–0.5080j)/T

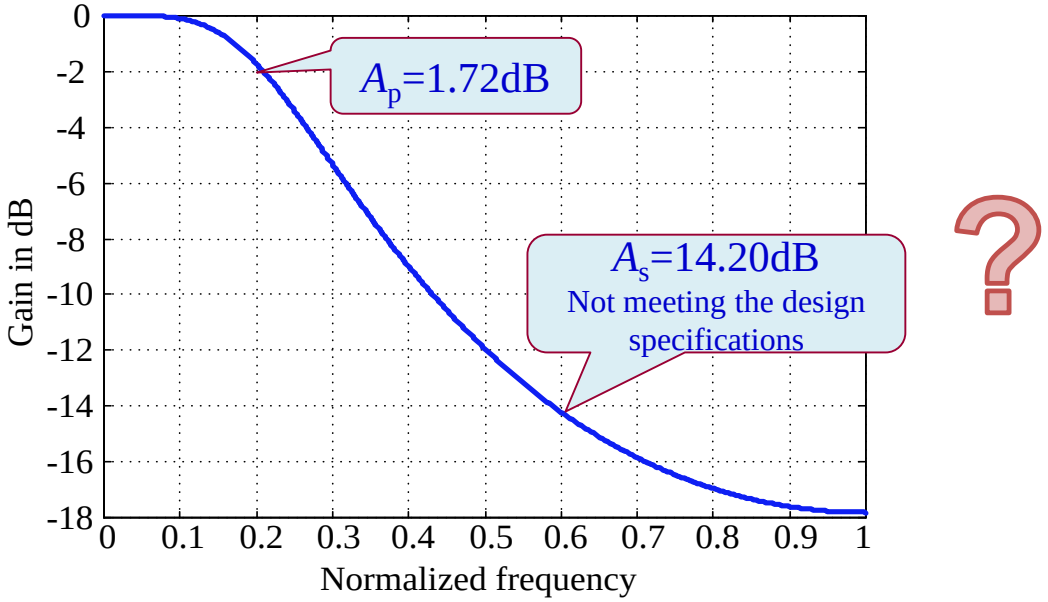
Use $\dfrac{1}{s-s_i} \longrightarrow \dfrac{1}{1-e^{s_i T}z^{-1}}$

The sampling interval T can be arbitrarily selected

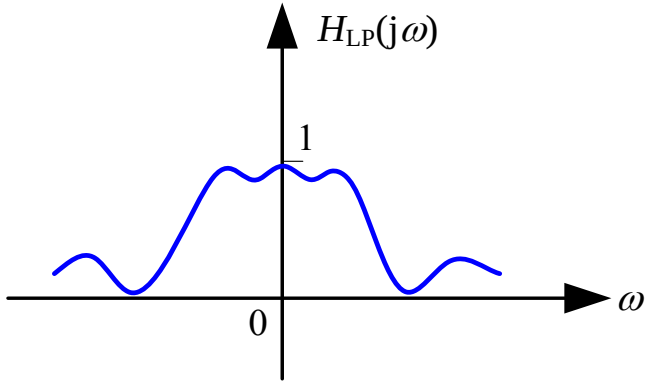
The system function of DF can be obtained as

$$H(z)=\frac{0.3104z^{-1}}{1-1.0514z^{-1}+0.3620z^{-2}}$$

[E.g] **Using BW type analog low-pass filter and impulse response invariant method to design and meet the specifications Ω_p=0.2π rad, Ω_s=0.6π rad, A_p≤2dB, A_s≥15dB digital low-pass filter.**



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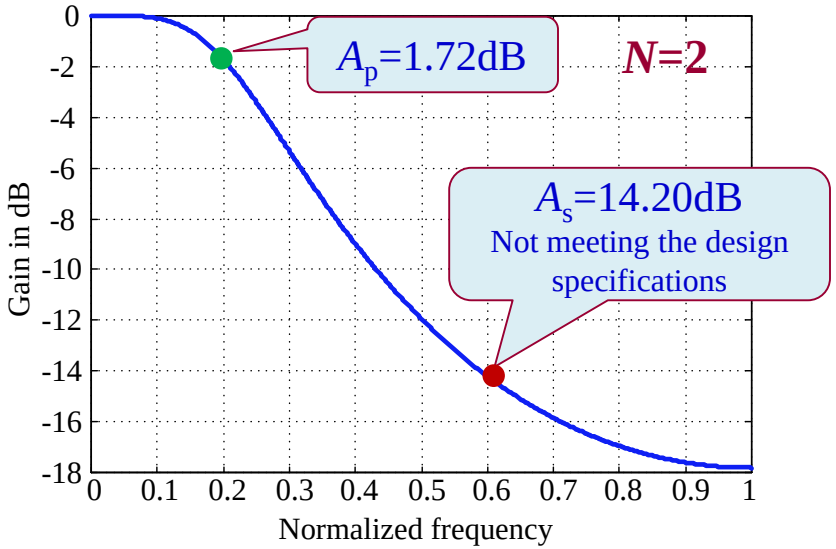


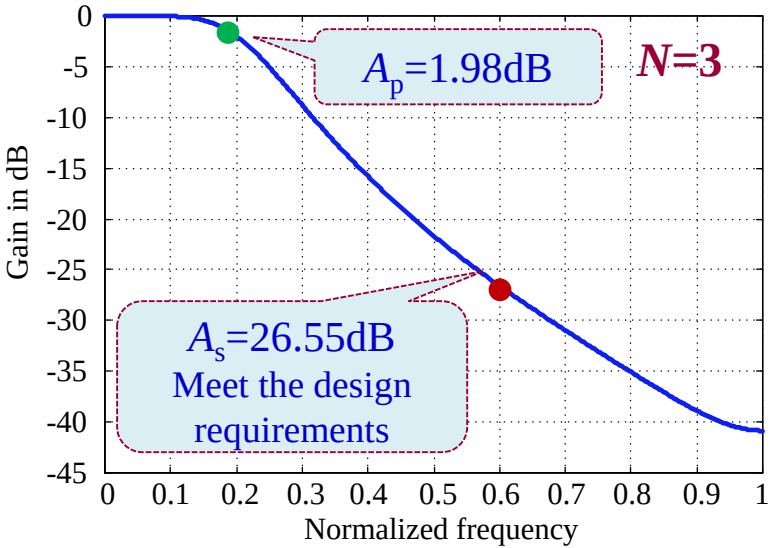
Practical problem: AF generally cannot meet the band limit conditions.

Impact: There is a certain degree of aliasing in the frequency response of DF.

How to solve it?

[E.g] **Using BW type analog low-pass filter and impulse response invariant method to design and meet the specifications Ω_p=0.2π rad, Ω_s=0.6π rad, A_p≤2dB, A_s≥15dB digital low-pass filter.**





Solution: Increase the stopband attenuation A_s of the DF to be designed

Convert analog filter to digital filter

Impulse Invariance

[numd,dend] = impinvar(num,den,Fs)

num,den：Coefficient vectors of AF numerator and denominator polynomials

Fs=1/T：sampling frequency

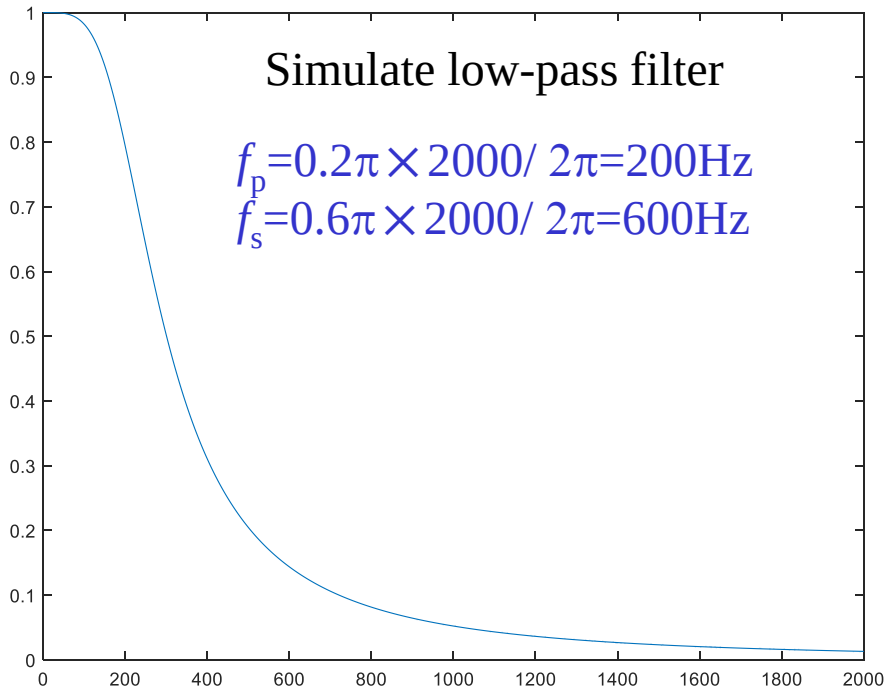
numd,dend：Coefficient vectors of DF numerator and denominator polynomials

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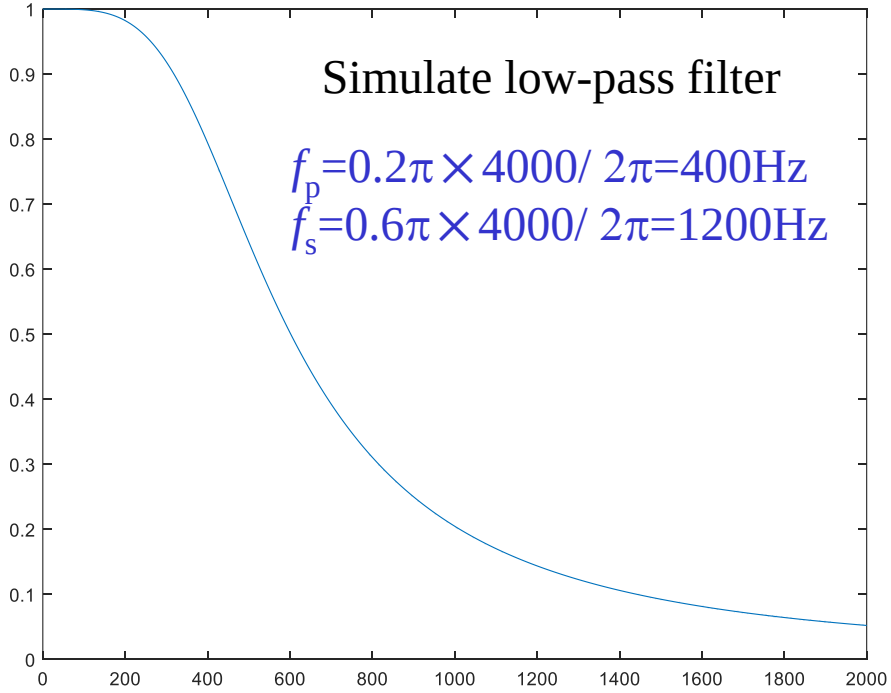
```
%Design DF BW low-pass filter
%DF BW LP specification
Wp=0.2*pi; Ws=0.6*pi; Ap=2; As=15;
Fs=1;   %Sampling frequency(Hz)
%Analog Butterworth specification
wp=Wp*Fs; ws=Ws*Fs;
%determine the order of AF filter
N=buttord(wp,ws,Ap,As,'s');
%determine the 3-db cutoff frequency of
BW filter from pass-band specification
wc=wp/(10^(0.1*Ap)-1)^(1/N/2);
%determine the AF-BW filter
 [numa,dena]=butter(N,wc,'s');
```

```
 %determine the DF filter
 [numd,dend]=impinvar(numa,dena,Fs);
%plot the frequency response
 w=linspace(0,pi,1024);
 h=freqz(numd,dend,w);
%normalize amplitude
 norm=max(abs(h));
 numd=numd/norm;
 plot(w/pi,20*log10(abs(h/norm)));
 xlabel('Normalized frequency');
 ylabel('Gain in dB');
 %computer Ap As of the designed filter
 w=[Wp Ws];
 h=freqz(numd,dend,w);
 fprintf('Ap= %.4f\n',-20*log10( abs(h(1))));
 fprintf('As= %.4f\n',-20*log10( abs(h(2))));
```

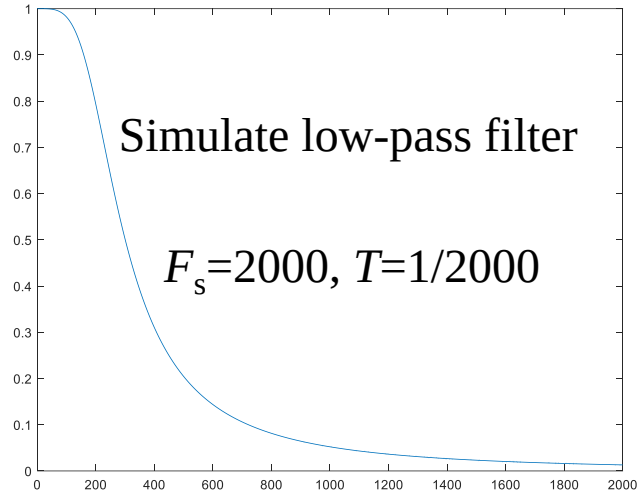
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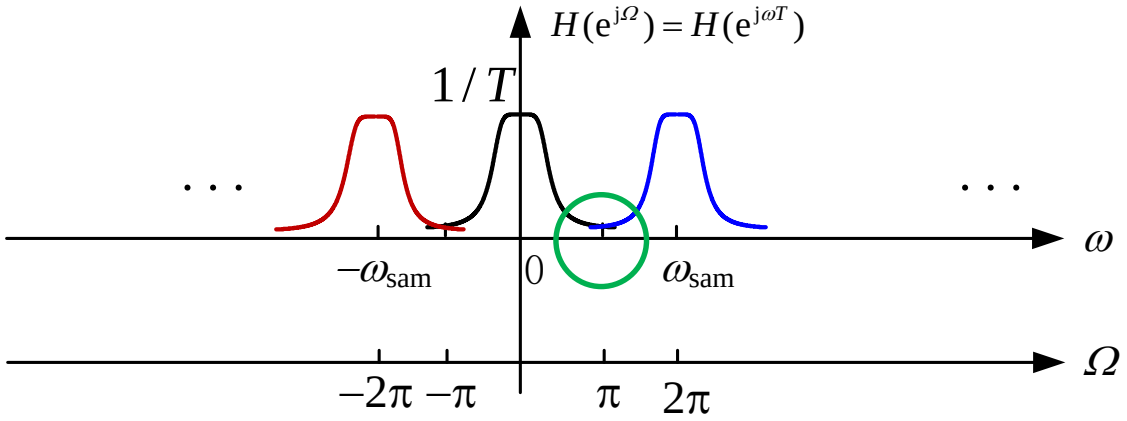


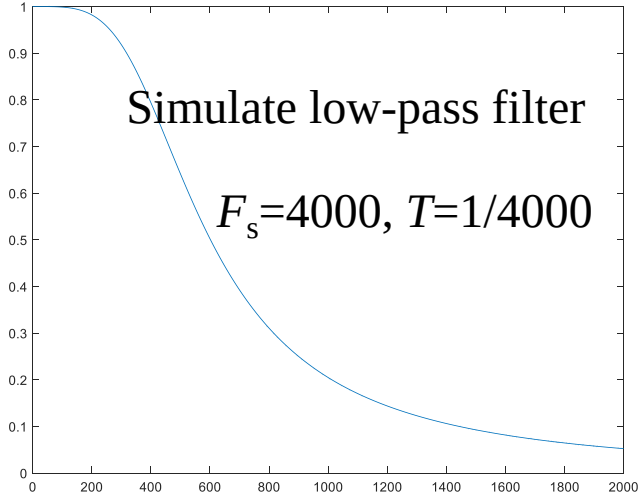
F_s=2000, T=1/2000

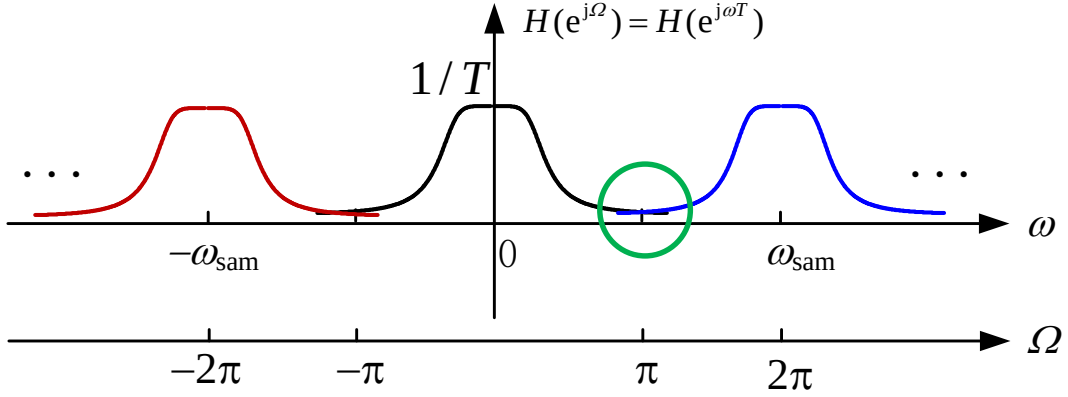


F_s=4000,T=1/4000









Taking different values of T results in the same aliasing effect

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.